Linear combination

Linear combination : A linear combination of the set of vectors $\{\boldsymbol{v}^{(1)}, \dots, \boldsymbol{v}^{(n)}\}$ is given by multiplying each vector by a corresponding scalar coefficient and adding the results

Scalars

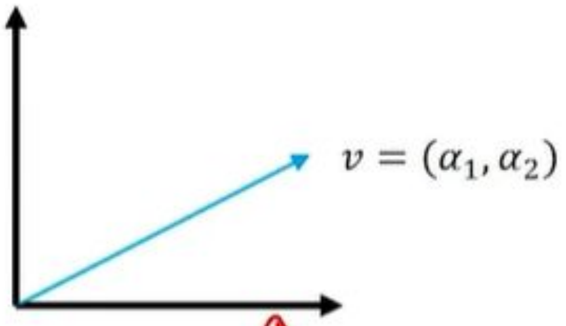
$$\sum_i \alpha_i \boldsymbol{v}^{(i)} = \alpha_1 v^{(1)} + \alpha_2 v^{(2)} \dots + \alpha_n v^{(n)}$$

Example : $\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\boldsymbol{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$

Linear Combination

$$\boldsymbol{v}_3 = v_1 + 2v_2 = \begin{bmatrix} 5 \\ 2 \\ 9 \end{bmatrix}$$

Span : The span of a set of vectors is the set of all vectors obtainable by a linear combination of the original vectors.



Whole of $\mathbb{R}^2$

Example : The span of the coordinate vectors $v_1 = (1,0)$, $v_2 = (0,1)$ is?

Ans : $\alpha_1 v_1 + \alpha_2 v_2 = (\alpha_1, \alpha_2)$